

BRAYDEN GOLDSTEIN-GELB

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Research Interests: Quantum algorithms and complexity theory, quantum error correction, mathematical structures in quantum computing, distributed quantum computing

EDUCATION

Brown University

Providence, RI

Sc.B Computer Science (Honors), Sc.B Mathematics

2021 – 2025

- GPA: 3.91/4.00, Magna Cum Laude
- Honors Thesis: Distributed Quantum Signal Processing. (External Advisor: Yuan Liu, NC State; Reader: Brenda Rubenstein, Brown; Department Advisor: James Tompkin, Brown)

RESEARCH EXPERIENCE

Postbaccalaureate Researcher, QUEST Lab

Remote

North Carolina State University

Jul '25 – Present

- Part-time continuation of research with Prof. Yuan Liu focusing on distributed quantum algorithms.

Undergraduate Researcher

Providence, RI

Brown University

Aug '24 – May '25

- Conducted honors thesis research on Distributed Quantum Signal Processing, extending QSP protocols to multi-processor quantum architectures via teledata and telegate methods while preserving asymptotic width and depth.
- Advised by Prof. Yuan Liu (NC State) and in collaboration with researchers from the Yale Quantum Institute, QuEra, and MIT. Manuscript in preparation.

Undergraduate Research Intern, SULI Program

Oak Ridge, Tennessee

Oak Ridge National Laboratory

May '23 – Jul '23

- Developed methods for solving constrained optimization problems using the Quantum Approximate Optimization Algorithm (QAOA)
- Mathematically proved convergence guarantees and conducted simulations to demonstrate strong performance.
- Supervised by Dr. Phillip Lotshaw.

Undergraduate Researcher, Brown Visual Computing Lab

Providence, RI

Brown University

May '22 – May '23

- **Decoupled Style Descriptors Project:** Built an interactive application for handwriting synthesis via a recurrent neural network.
- **Augmented Reality Label Visualization Project:** Designed and implemented real-time algorithms for adapting augmented reality content to dynamic environments using Unity shaders.
- Advised by Prof. James Tompkin.

PUBLICATIONS AND PRESENTATIONS

PAPERS

- **Brayden Goldstein-Gelb**, Kun Liu, John Martyn, Yongshan Ding, Yuan Liu. COMPAS: A Distributed Multi-Party SWAP Test for Parallel Quantum Algorithms. *In Submission*
- **Brayden Goldstein-Gelb**, Phillip C. Lotshaw. Convergence guarantee for linearly-constrained combinatorial optimization with a quantum alternating operator ansatz. *Under review*, ACM TQC (submitted Dec 2024); preprint available at arXiv:2409.18829.

POSTERS

- Ashley Kwon, Yuanbo Li, Eva Schiller, **Brayden Goldstein-Gelb**, James Tompkin. Environment Adaptive AR Label Visualization. Brown University Undergraduate Research Symposium, April 2023. Poster presentation.

COURSEWORK

- **Computer Science:** An Algorithmist's Toolkit (grad.), Design and Analysis of Algorithms, Theory of Computation, Algorithmic Machine Learning, Applied Cryptography, Systems, Formal Proof and Verification, Computer Vision
- **Mathematics:** Four semesters of Algebra (two undergrad, two grad), Real Analysis, Complex Analysis, Graph Theory, Probabilities in Quantum Mechanics, Information Theory, Number Theory, Optimization and Stochastic Calculus, Calculus on Manifolds, Statistical Inference, Linear Algebra
- **Physics/Engineering:** Quantum Mechanics, Quantum Information, Dynamics and Vibrations, Mechanics

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, HTML/CSS, JavaScript, C/C++, Julia, Pyret
- **Technologies:** Qiskit, Git, NumPy, Sklearn, Pandas, TensorFlow, PyTorch, SLURM/HPC, D3, Node.js

TEACHING

Grader, Linear Algebra with Theory Spring '25

- Graded problem sets for a proof-based linear algebra course

Teaching Assistant, Theory of Computation Fall '24

- Created problem sets and example solutions, assisted in grading, held weekly office hours

Online Tutor, Wyzant Inc. May '21 – Aug '23

- Over 350 hours of tutoring experience, working with 50+ students ranging from middle schoolers to executives.
- Maintained a five-star rating, helping students build confidence and mastery in Python, Java, Calculus, and Algebra.

PROJECTS

Hackathon Projects, MIT iQuHack, Brown Quantum Hackathon, Hack@Brown Spring '23 – Fall '24

- Won 2nd Place at Brown Quantum Hackathon
- Completed QuEra challenge at MIT's iQuHack quantum hackathon
- Created a space-themed game for Hack@Brown

Zero-Knowledge Proof for Graph 3-Coloring, Final Project, Applied Cryptography Spring '24

- Implemented an interactive zero-knowledge protocol to prove graph 3-colorability without revealing the coloring. *View project report.*
- Built a terminal-based application with verifier/prover roles communicating over TCP sockets.

Quantum Mechanics Proofs Library, Final Project, Formal Proof and Verification Fall '22

- Created a library in the Lean Theorem Prover to model quantum particles, measurements, and multi-particle states.
- Demonstrated the library's capabilities by constructing a formal proof of the no-cloning theorem.

More projects, additional details, and demos available at brayden-gg.github.io/projects.

AWARDS

- **Sigma Xi**, Scientific Research Honor Society
- **Audience Favorite Award**, Brown Undergraduate Research Symposium